

Drug acting on the central nervous system

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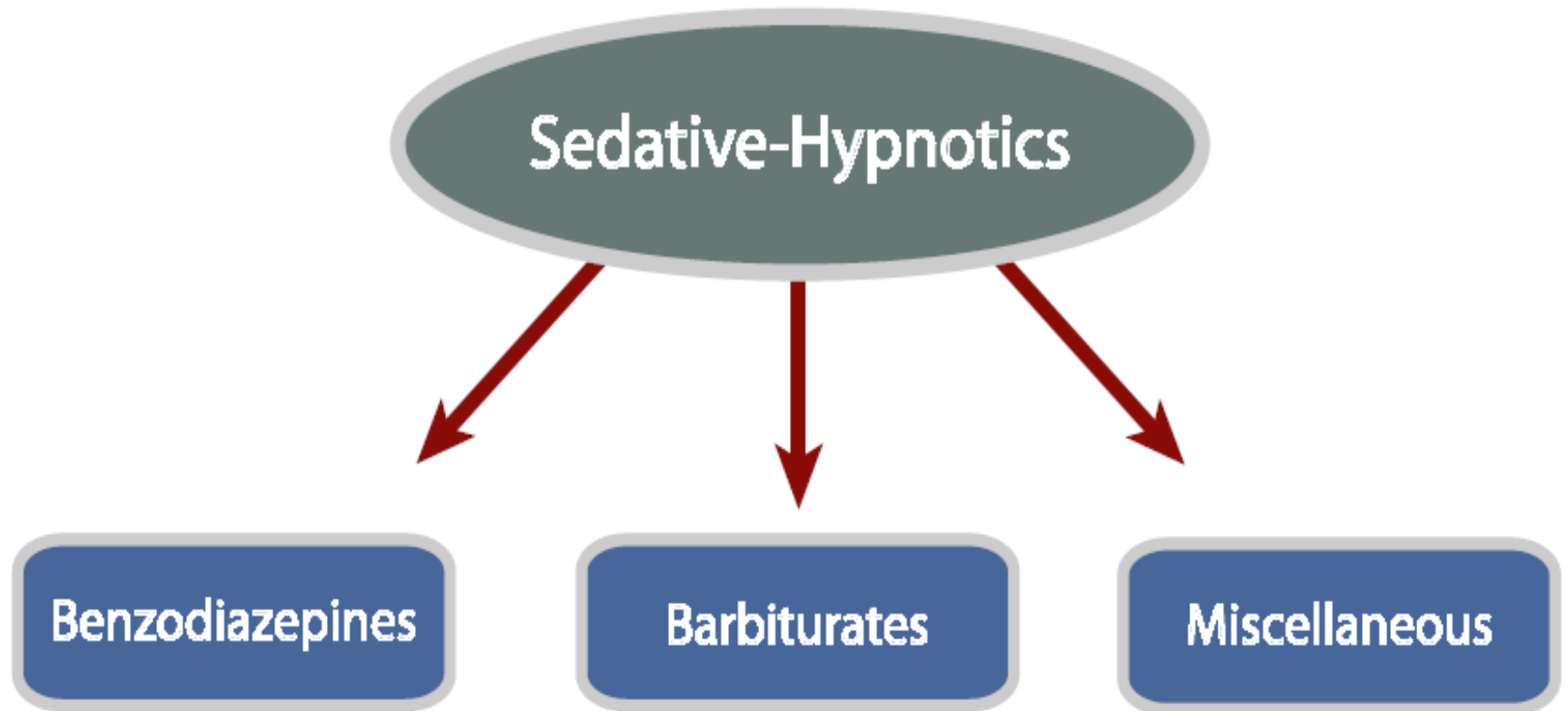
- Drugs affecting the CNS provide
 - Anesthesia, treats psychiatric disorders, relieve anxiety, provide sleep or sedation, prevent epileptic seizure, and suppress movement
 - Lipid soluble non-ionized molecules pass readily into the CNS (BBB)

1. Sedative and hypnotics

- Sedative drugs provide a quieting effect accompanied by relaxation and rest but not necessarily sleep
- No sharp distinction between the two types
- Also employed as anticonvulsant, muscle relaxant and anti-anxiety

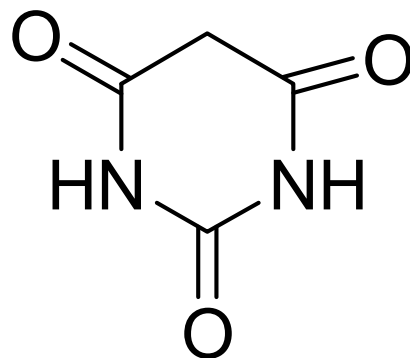
Classification of Sedative-Hypnotics

- The sedative-hypnotics may be divided into three major groups:
 - ❑ Barbiturates
 - ❑ Benzodiazepines
 - ❑ Miscellaneous



1. Barbiturate

- Barbituric acid was synthesized in 1864 by a German chemist, Adolph von Baeyer from a reaction of urea with malonic acid, a chemical found in apples.

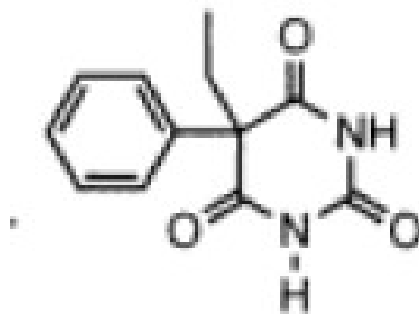


- But was not used as a drug

- In 1903, Fischer and von Mering were the first to synthesize a therapeutically active "barbiturate" by substituting two ethyl groups for two hydrogens attached to carbon no. 5.
 - When they administered this new barbiturate to human subjects, the compound was found to induce sleep.
 - Diethyl barbituric Acid (Veronal) was a popular drug.
 - It allowed sleep at night, and even caused drowsiness and relaxation when taken during the day.
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- Are 5, 5 disubstituted derivatives of barbituric acid

- Problem was, slow acting and was very slow to wear off due to slow metabolism.
- So those who took it may wind up sleeping a day and a half!

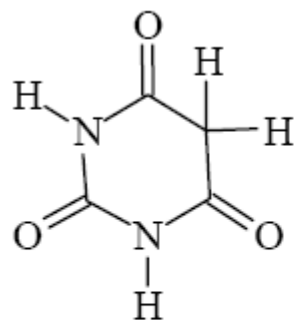
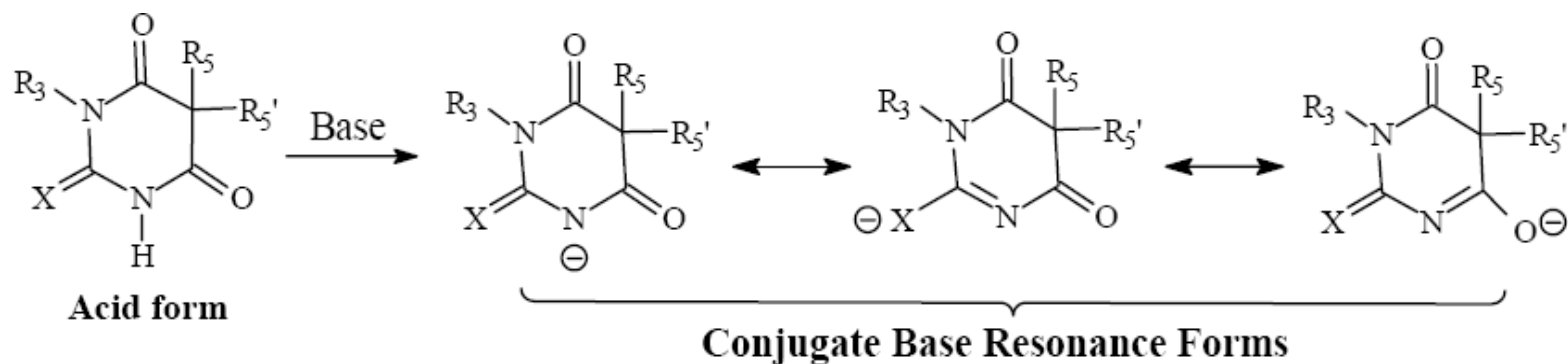
- In 1912, two independent teams of chemists synthesized what became known as Luminal, or phenobarbital.
- In addition to the excellent hypnotic action, the compound was found to have anticonvulsant activity.
- The compound, given twice daily, kept seizures under control.
- And the hypnotic effects could be counteracted by administering amphetamines without affecting the anticonvulsant properties.



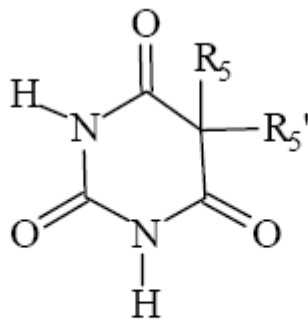
- Subsequent research on barbiturates began to understand that the **lack of drug activity in barbituric acid** and **the slow acting effect of Veronal** were caused by the same property
 - Negligible or slow absorption through GIT and limited penetration through BBB

- Based on the above findings several other barbiturates were synthesized (**by trial and error**) and their biological activity was tested.

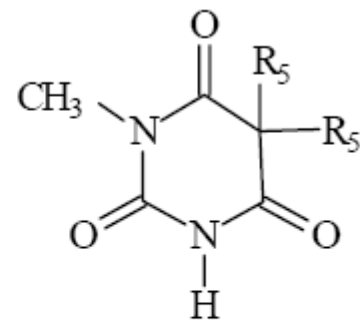
- Consideration of the structure of 5,5-disubstituted barbituric acids reveals their acidic character.
- Those without methyl substituent on the nitrogen have $pK_a \sim$ of about 7.6: those with methyl substituent have pK_a about 8.4.
- The free acids have poor water solubility and good lipid solubility (the latter largely a function of the **two hydrocarbon substituent on the 5 position**. although in the **2 thiobarbiturates** the **sulfur atom increases lipid solubility**).



barbituric acid: pKa 4.12

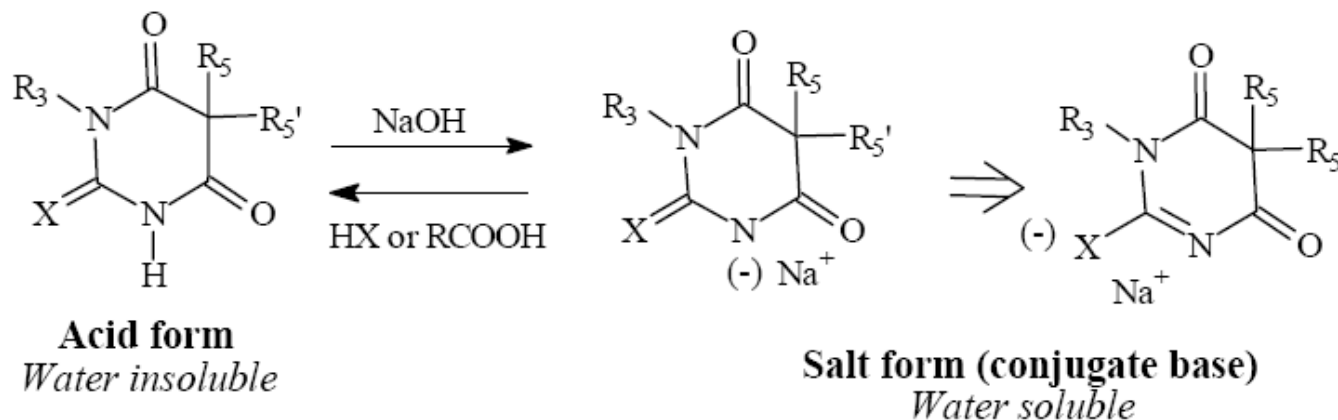


**5,5'-Disubstituted
barbituric acid: pKa 6.5-8**

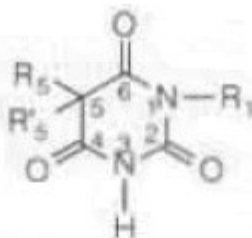


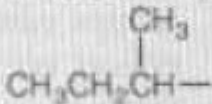
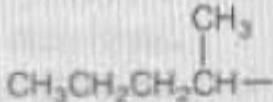
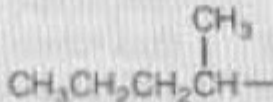
**3,5,5'-Trisubstituted
barbituric acid: pKa > 8**

- Sodium salts of the barbiturates are readily prepared and are water soluble
- Their aqueous solutions generate an alkaline PH
- A classic incompatibility is the addition of an agent with an acidic pH in solution which results in formation and precipitation of the free water insoluble disubstituted barbituric acid.

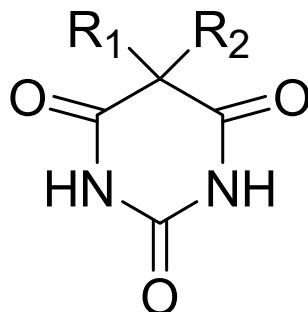


General Structure



Generic Name Proprietary Name	Substituents			Sedative Dose (mg)	Hypnotic Dose (mg)	Usual Onset of Action (min)
R ₅	R' ₅	R ₁				
A. Long Duration of Action (more than 6 hours)						
Mephobarbital, USP <i>Mebaral</i>	C ₂ H ₅		CH ₃	30-100 ^d	100	30-60
Phenobarbital, USP <i>Luminal</i>	C ₂ H ₅		H	15-30 ^d	100	20-40
B. Intermediate Duration of Action (3 to 6 hours)						
Amobarbital, USP <i>Amytal</i>	CH ₃ CH ₂ —	(CH ₃) ₂ CHCH ₂ CH ₂ —	H	20-40	100	20-30
Butobarbital Sodium, USP <i>Butisol Sodium</i>	CH ₃ CH ₂ —		H	15-30	100	20-30
C. Short Duration of Action (less than 3 hours)						
Pentobarbital Sodium, USP <i>Nembutal Sodium</i>	CH ₃ CH ₂ —		H	30	100	20-30
Secobarbital, USP <i>Seconal</i>	CH ₂ =CHCH ₂ —		H	15-30	100	20-30

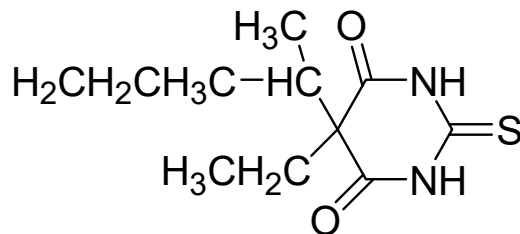
SAR



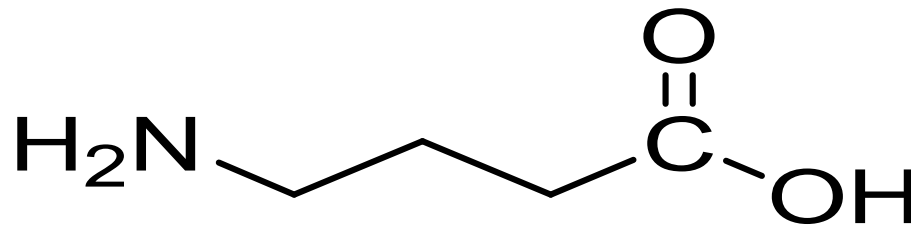
- Hypnotic activity increases with lipid solubility until the total number of carbon atoms at both C-5 substituents is between 6 and 10.
- Further increase in the sum of the number of carbon atoms decreases hypnotic activity despite increased lipophilicity, indicating that lipophilicity must remain within certain limits.
- Within the same series, **the branched chain isomer generally** has greater lipid solubility, hypnotic activity, and shorter duration of action than the straight chain isomer.

- Within the same series the **unsaturated allyl, alkenyl, and alkynyl** derivatives are **more hypnotic** than the saturated analogs with the same number of carbon atoms.
- Conversion of a 5, 5-disubstituted barbituric acid to a 1, 5, 5-trisubstituted analog does not result in a significant change in hypnotic activity.

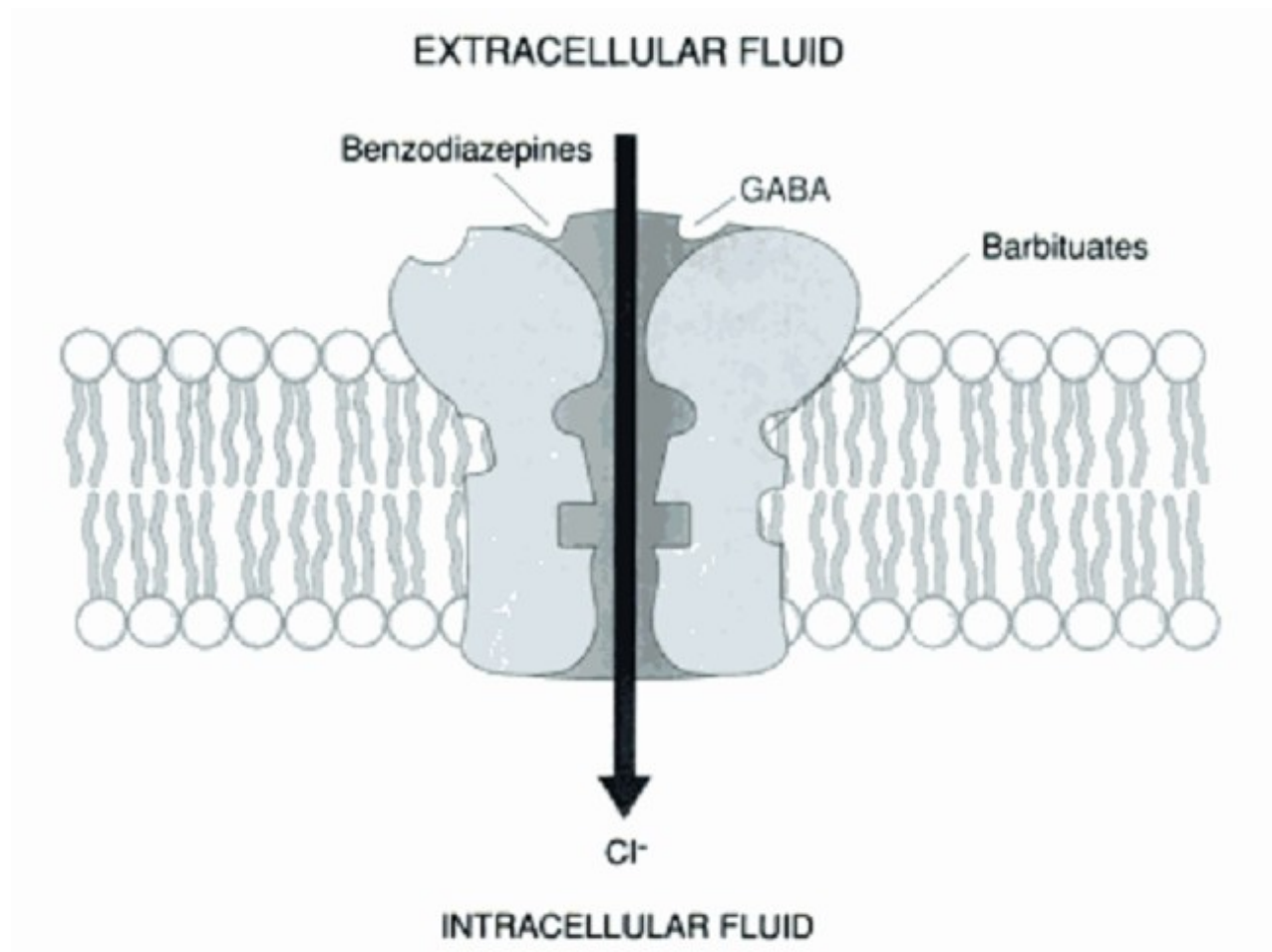
- Introduction of polar substituents (OH, NH,, COOH, CO, RNH, SO, H) into an aromatic moiety at the 5-position decreases lipid solubility and potency.
- Replacement of the oxygen at C-2 by a sulfur atom results in **faster onset** but **shorter duration of hypnotic activity**.



- MOA- prolong the inhibitory action of GABA
- Barbiturates bind to GABA receptors on the allosteric site.



GABA-γ amino butyric acid



Metabolism of barbiturates

- There are four primary metabolic pathways for barbiturates

1. Oxidation of substituents attached to C5

- It is the most important pathway of metabolism for the barbiturates.
- The oxidative processes may yield alcohols, ketones, and carboxylic acids.

2. N-Dealkylation (N-demethylation)

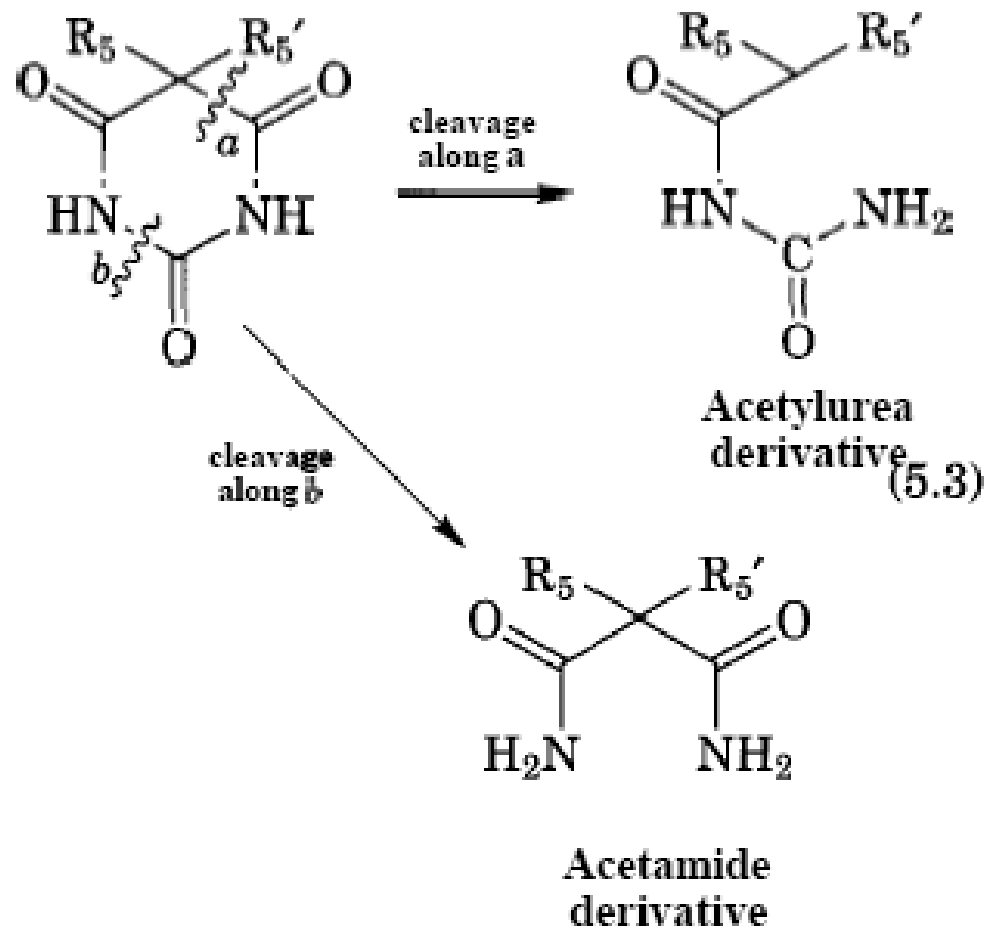
- It is an important metabolic pathway for N-substituted barbiturates.
- Example mephobarbital (1-methyl-5-ethyl-5-phenylbarbituric acid) is metabolized to phenobarbital (5-ethyl-5-phenylbarbituric acid), which is subject to further metabolic processes.

3. Desulfurization of 2-thiobarbiturates

- ❑ Is a common metabolic process.
- ❑ Example, pentobarbital [5-ethyl-5-(1-methylbutyl) barbituric acid] is one of the metabolic products of thiopental [5-ethyl-5-(1-methylbutyl) - 2-thiobarbituric acid].

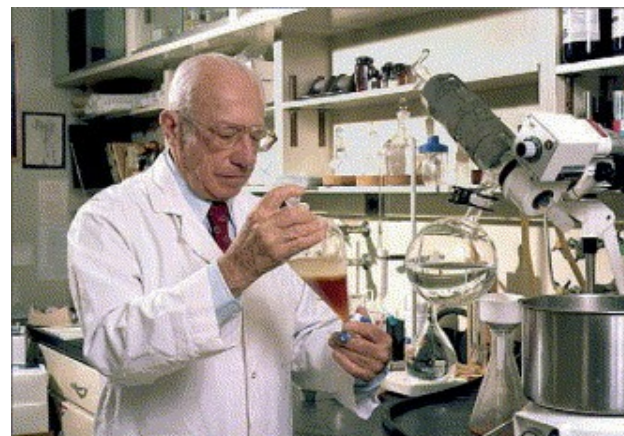
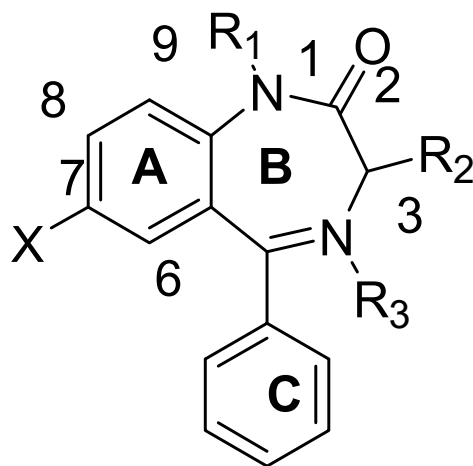
4. Ring scission of the barbituric ring

- ❑ Leads to the formation of acetamides or acetyl urea derivatives.

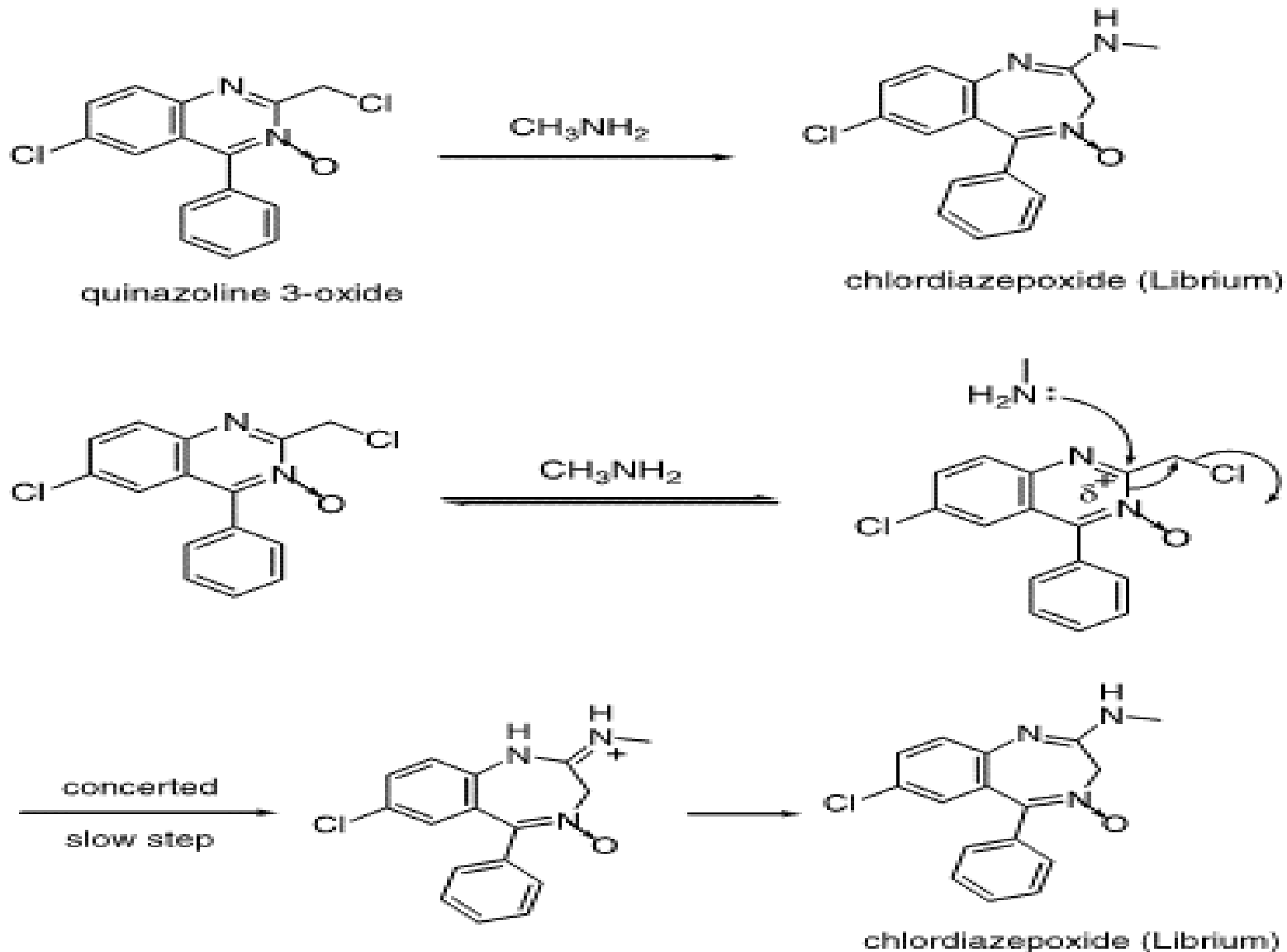


ii. Benzodiazepines

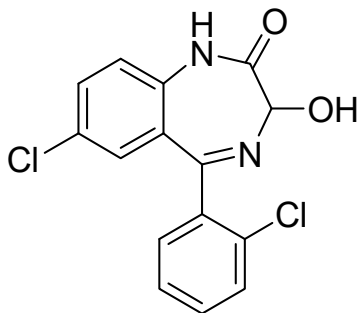
- **Benzodiazepine** refers to portion of the structure of **benzene** fused to seven membrane **diazepine** ring



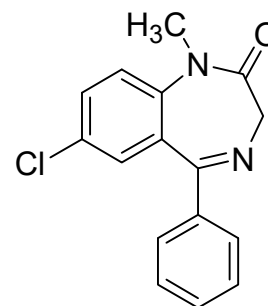
Sternbach BZD synthesis...



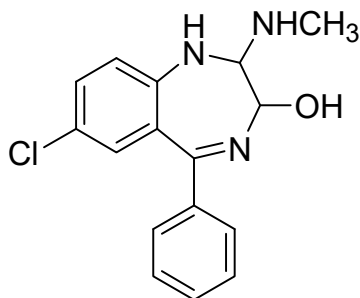
- Substituents at position marked R1, R2, R3 and X result in different drugs



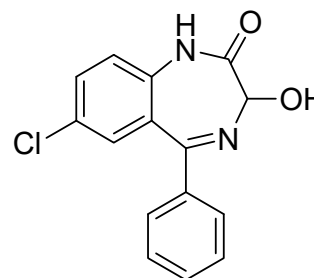
Chlordiazepam



Diazepam

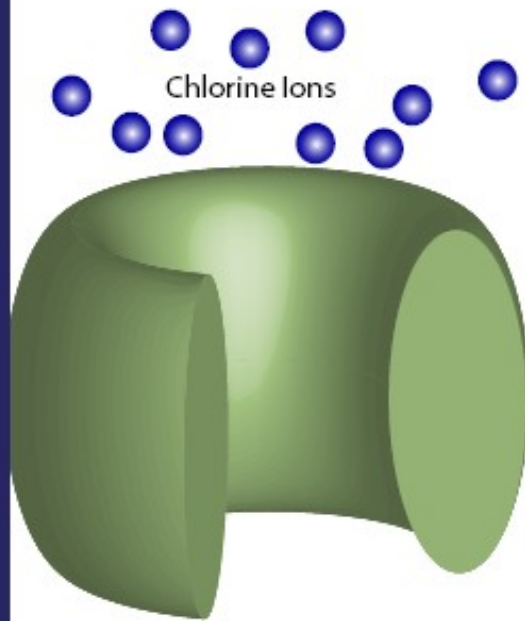


Lorazepam

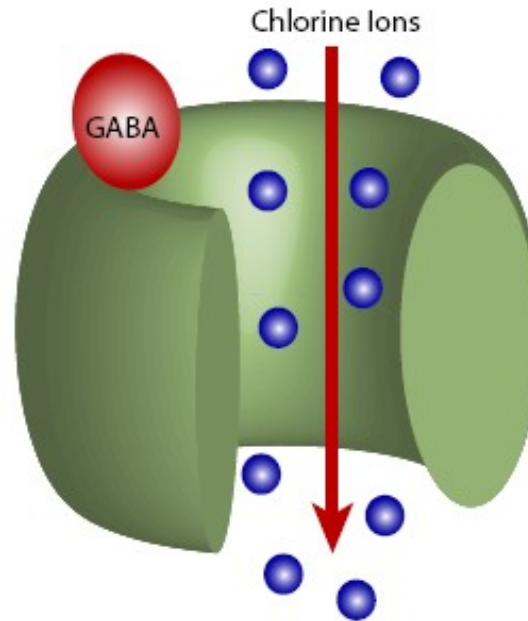


Oxazepam

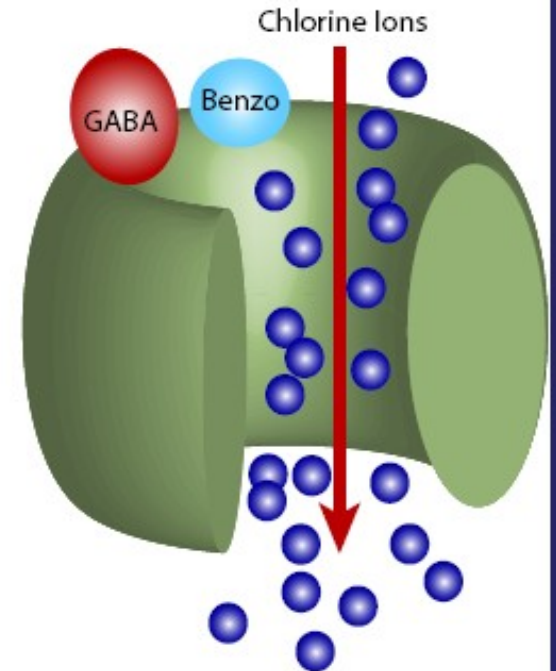
Extracellular Side of GABA_A Receptor



1. No chlorine influx because GABA has is not bound to receptor



2. Normal chlorine influx resulting from GABA binding to receptor



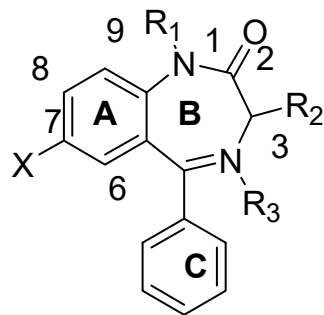
3. Increased chlorine influx resulting from both GABA and benzodiazepine binding to receptor

Intracellular Side of GABA_A Receptor

- The pharmacological effects of **benzodiazepines** are outlined as follows:
 - ❑ They produce sedation, hypnosis, decreased anxiety, muscle relaxation, and anticonvulsant activity.
 - ❑ Benzodiazepines are **not general neuronal depressants** as are barbiturates.
 - ❑ Benzodiazepines have similar pharmacologic profiles, but the drugs differ in selectivity. The clinical usefulness of benzodiazepine drugs varies accordingly.
 - ❑ Benzodiazepines enhance GABAergic transmission in all CNS structures.

- Benzodiazepines show **fewer tendencies to tolerance** and dependency than other older sedative-hypnotic drugs, especially barbiturates.
- Also, benzodiazepines produce less abuse potential.
- Benzodiazepines are safer in overdose especially compared with barbiturates.
- Benzodiazepines produce fewer drug interactions because they do not induce hepatic microsomal enzymes

SAR of benzodiazepines



- Saturation of the 4, 5 double bond or shift to 3, 4-position decreases activity
- Position 6, 8, and 9 should not be substituted
- The 2-carbonyl function is optimal for activity as is the nitrogen atom at position 1.
 - The N substitution should be small

- A phenyl group at the C5 promotes activity
 - Ortho or di-ortho substitution with **electron attracting Substituents** increases activity.
 - **Para Substituents** decrease activity
- **Electron attracting Substituents** at C7 is required for activity
 - The more electrons attracting, the higher the activity
- Hydroxyl group at C3 is important pharmacokinetically
 - Compounds without the hydroxyl group are non-polar and have longer half-life
 - Those with hydroxyl group are polar and readily excreted